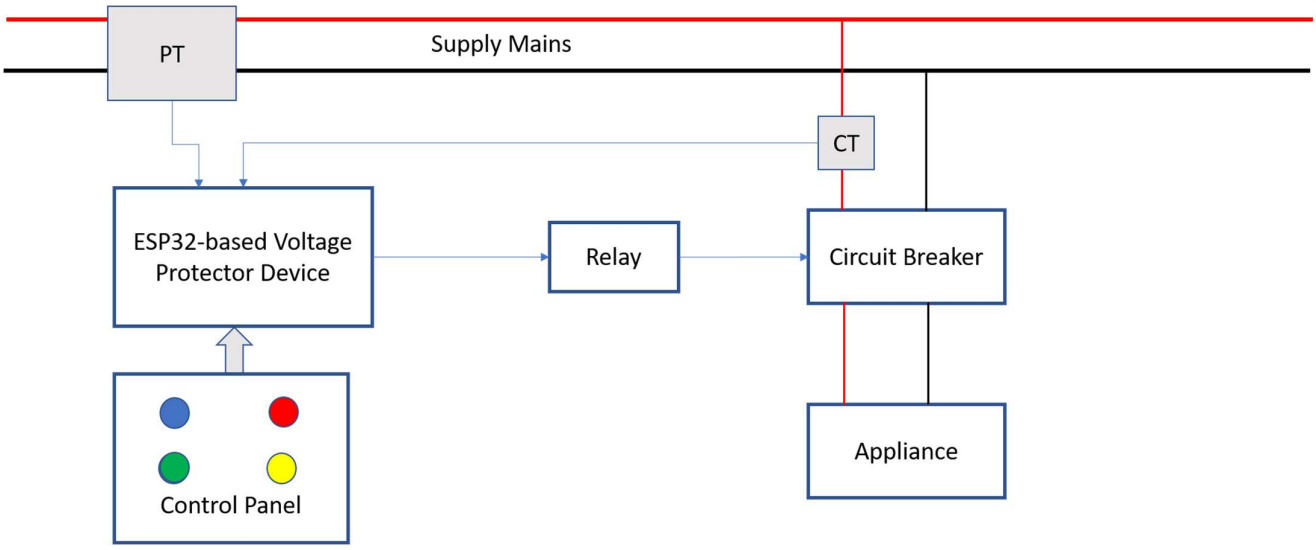


Fig. 01: Project layout of an ESP32-based Voltage Protector Device

My Thoughts on the Project:

1. The CT and PT provide the real-time true-rms voltage and true-rms current update. These analog values are converted into digital values (ADC) inside the ESP32-based device.
2. If a voltage surge happens in the grid/supply mains, then the voltage rises very rapidly. So, this device should check the deviation of the present voltage from the last one, and compare it with a predefined threshold to determine whether there is a voltage spike. If a spike is detected, the relay operates immediately and the circuit breaker trips, thus the appliance is disconnected from the supply mains.
3. This device also should also check whether if the voltage remains above the lower threshold voltage, because a voltage lower than the minimum voltage means that there might be an overcurrent fault in the supply mains (which is normally caused due to excessive load). In that case, the device should also trip the circuit breaker.
4. The CT here measures how much current the appliance is taking itself. If it exceeds the maximum rating of the appliance, then the circuit breaker should trip to save it from damage.

The Tasks I have Completed in the Simulation Project:

1. I have simulated my whole project in “Wokwi” web-based simulation software. There, it lacks actual power grid integration. So, I have taken two potentiometer whose output signal mimics the rms voltage and current output from PT and CT. If the voltage falls below the minimum threshold or rise above the maximum threshold, then the relay operates, so does the circuit breaker.
2. The settings of the device can be changed manually by pressing some pushbuttons. The thresholds can be changed as users wants.
3. Once the device detects a fault, the appliance is disconnected from the supply. To take the device back to its running condition the ‘reset’ button needs to be pressed after manual connection of the appliance after the faults are cleared.
4. I also tried to capture any voltage surge in the grid by checking the voltage difference between two consecutive timestamps. If that is too high, that means a voltage surge is happened in the grid. In that case the appliance should be disconnected.